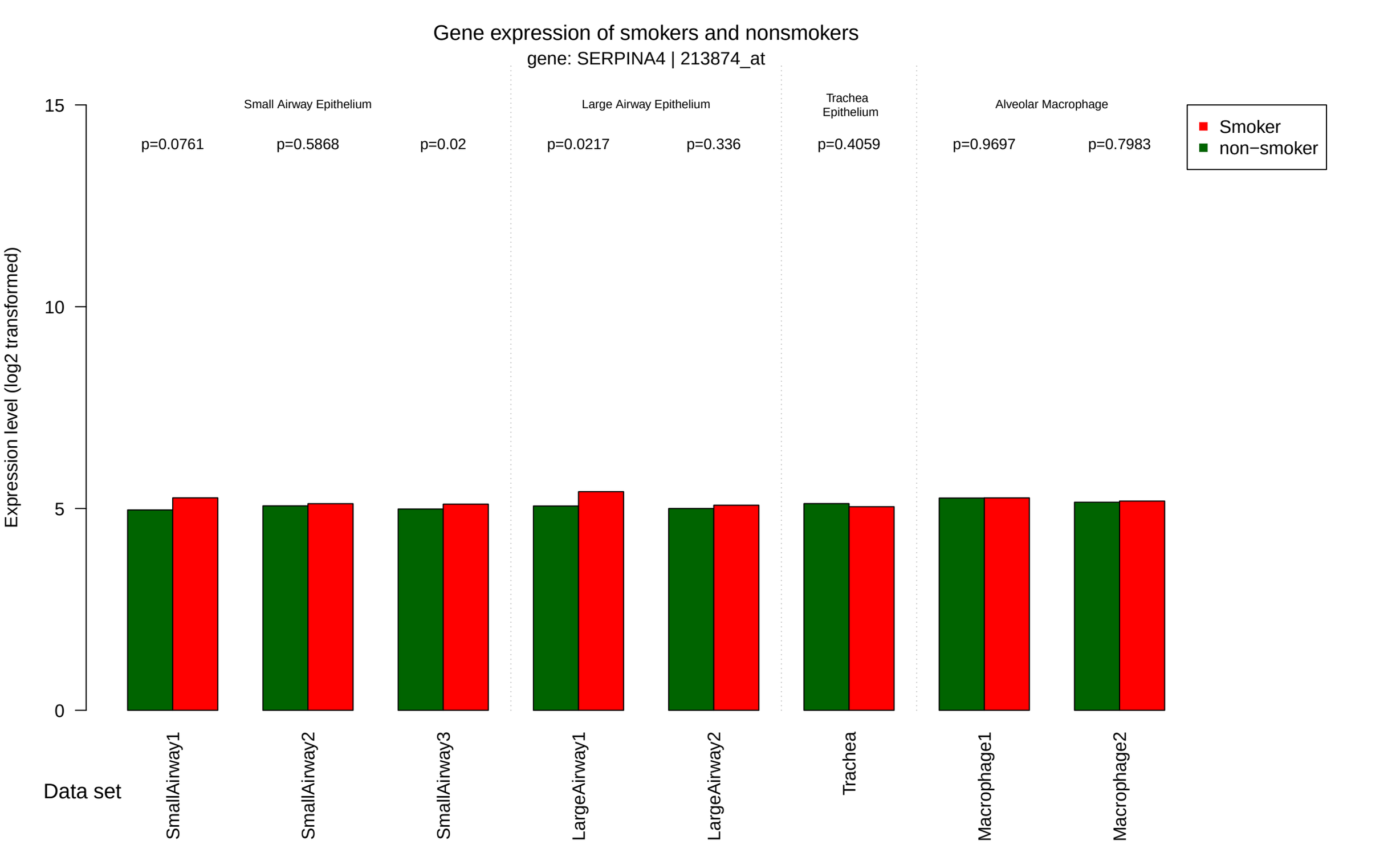
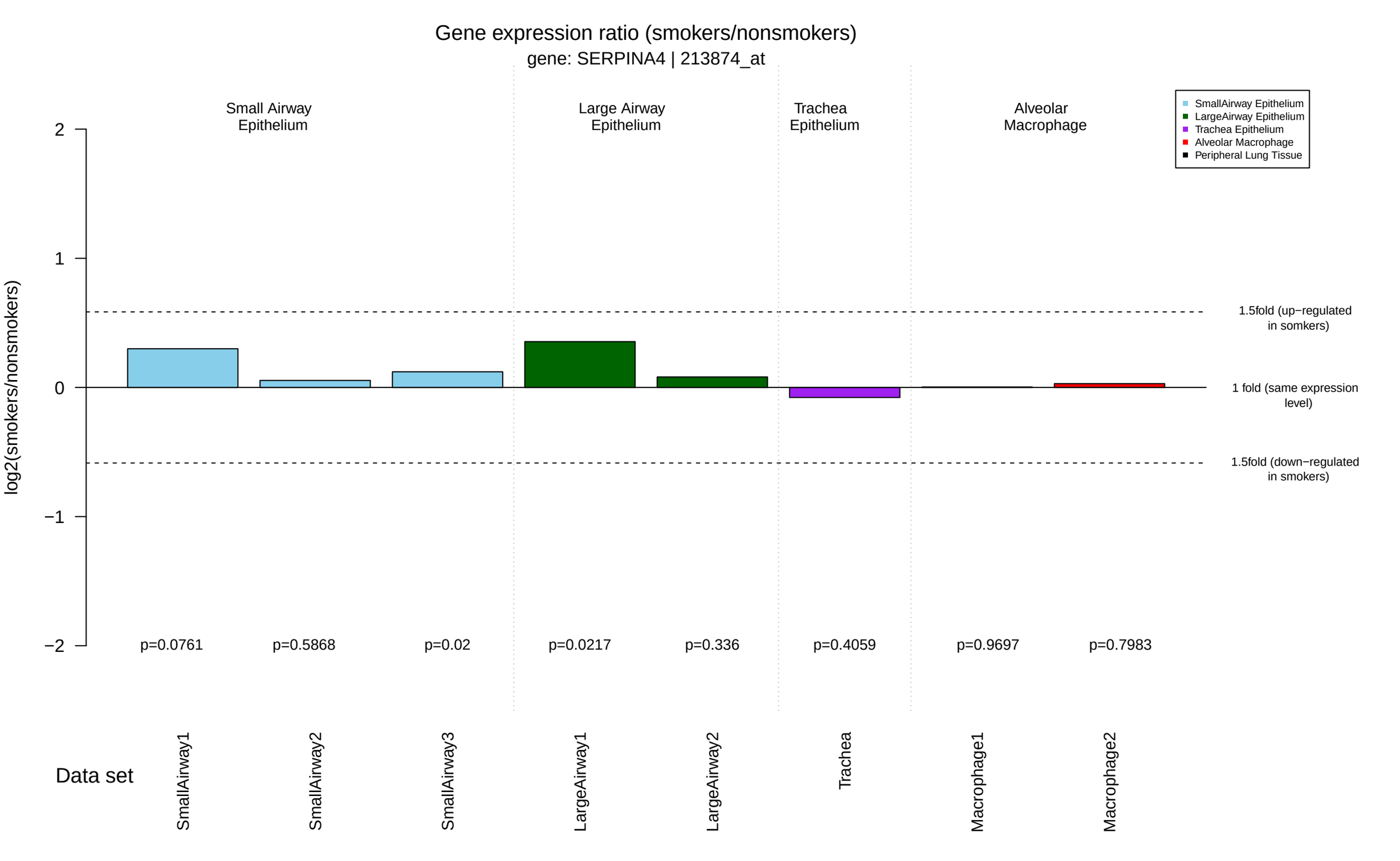
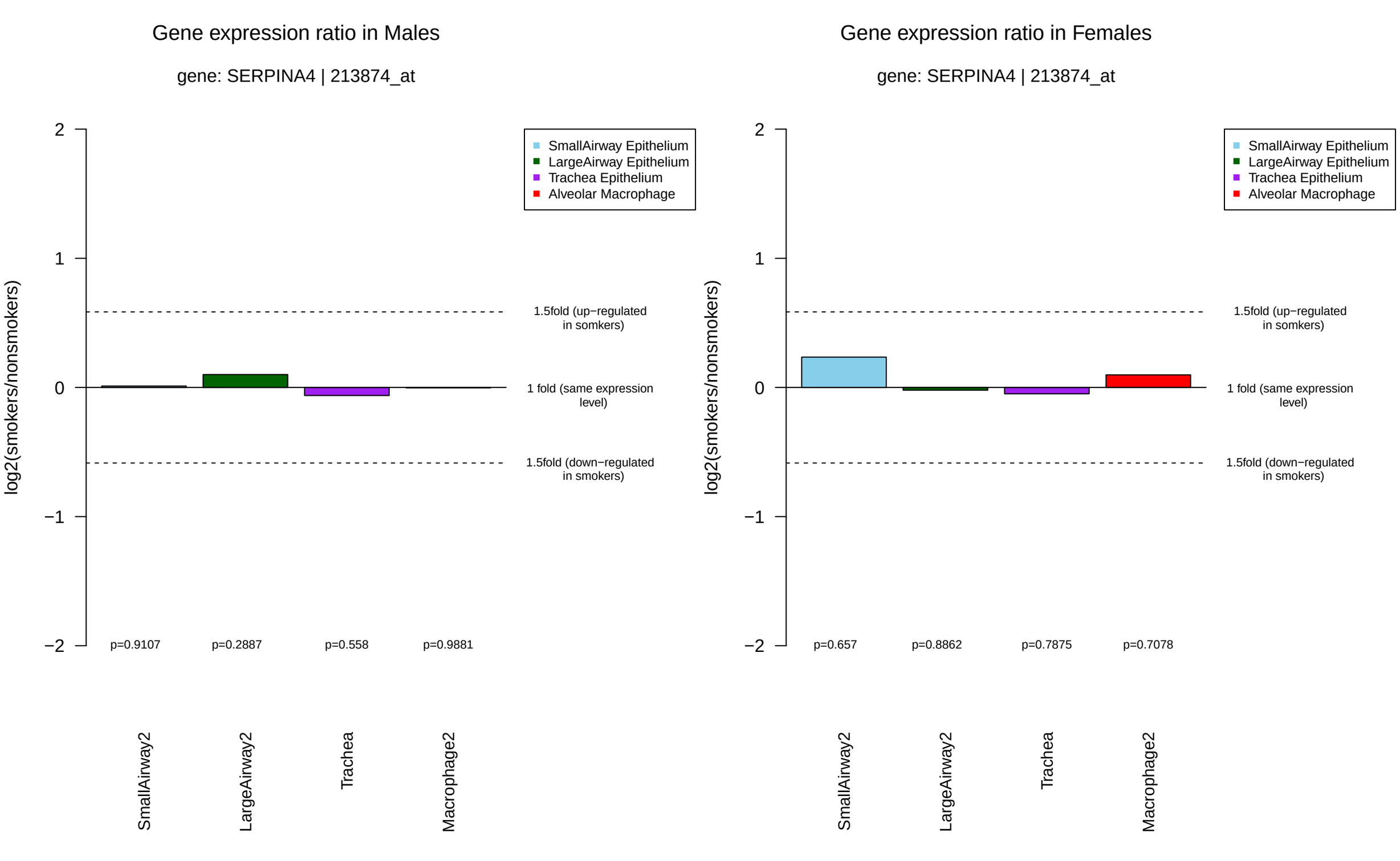
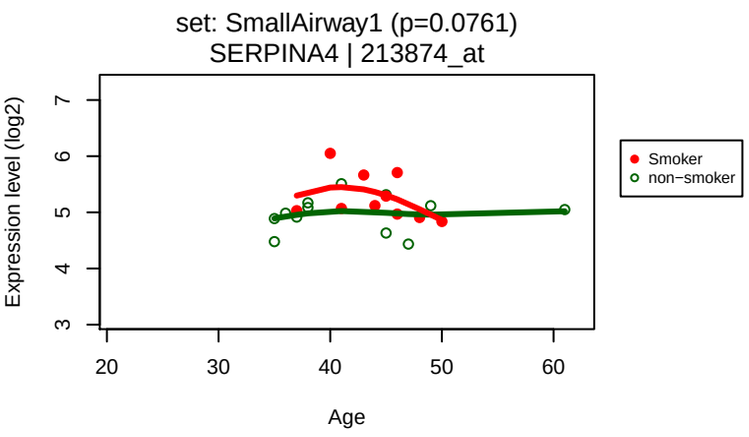
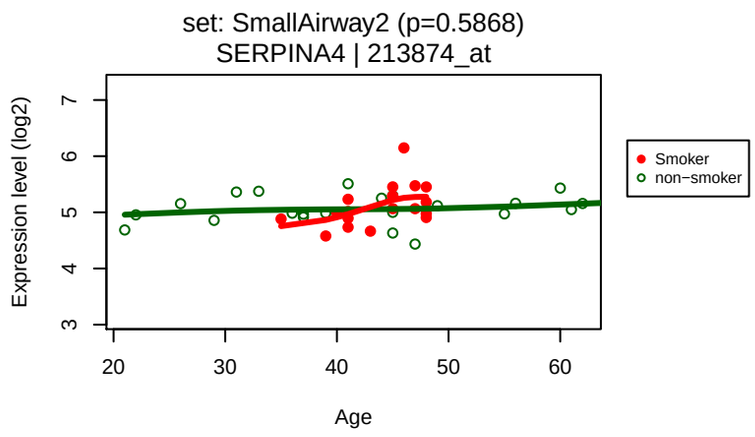
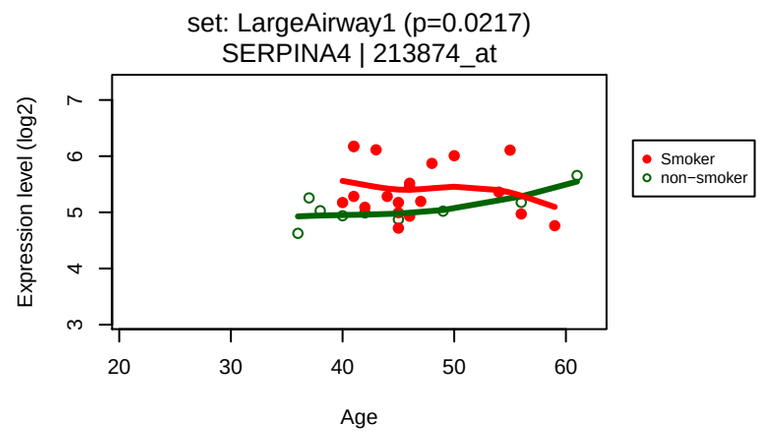

Gene expression of smokers and nonsmokers
gene: SERPINA4 | 213874_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0761
p=0.5868
p=0.02
p=0.0217
p=0.336
p=0.4059
p=0.9697
p=0.7983
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

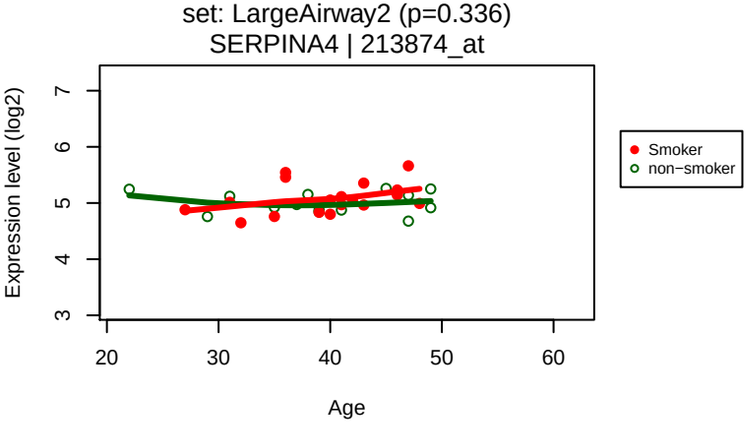

Gene expression ratio (smokers/nonsmokers)
gene: SERPINA4 | 213874_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0761
p=0.5868
p=0.02
p=0.0217
p=0.336
p=0.4059
p=0.9697
p=0.7983
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

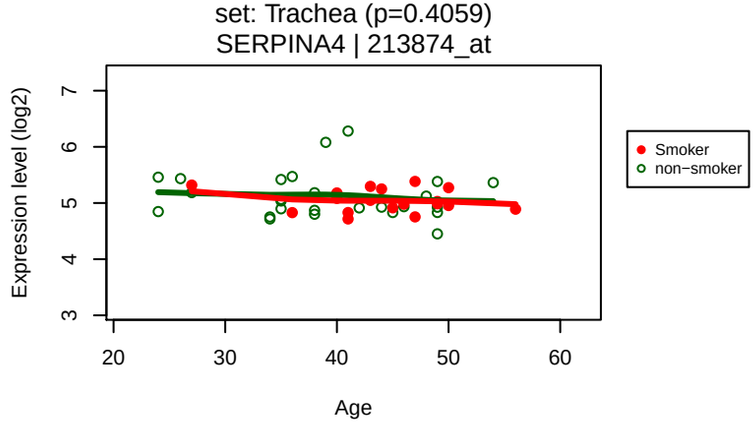

Gene expression ratio in Males
gene: SERPINA4 | 213874_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9107
p=0.2887
p=0.558
p=0.9881
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SERPINA4 | 213874_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.657
p=0.8862
p=0.7875
p=0.7078
SmallAirway2
LargeAirway2
Trachea
Macrophage2

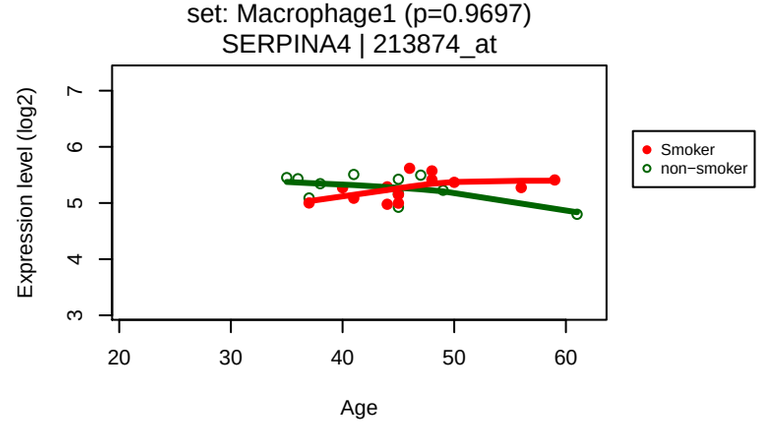

set: SmallAirway1 (p=0.0761)
SERPINA4 | 213874_at
Expression level (log2)
Age
Smoker
non−smoker

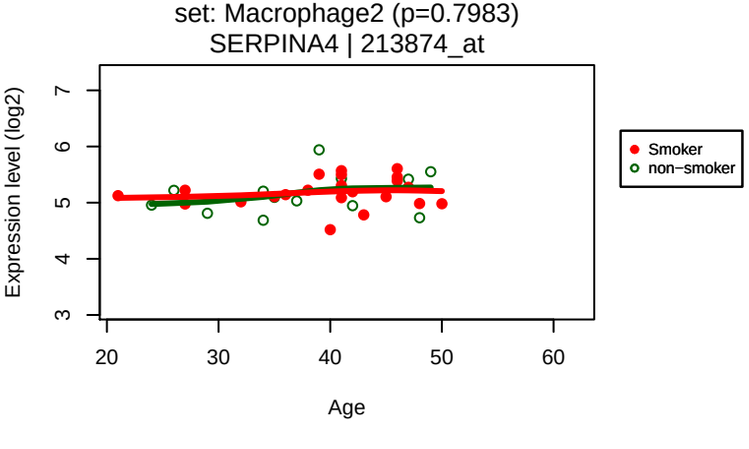

set: SmallAirway2 (p=0.5868)
SERPINA4 | 213874_at
Expression level (log2)
Age
Smoker
non−smoker

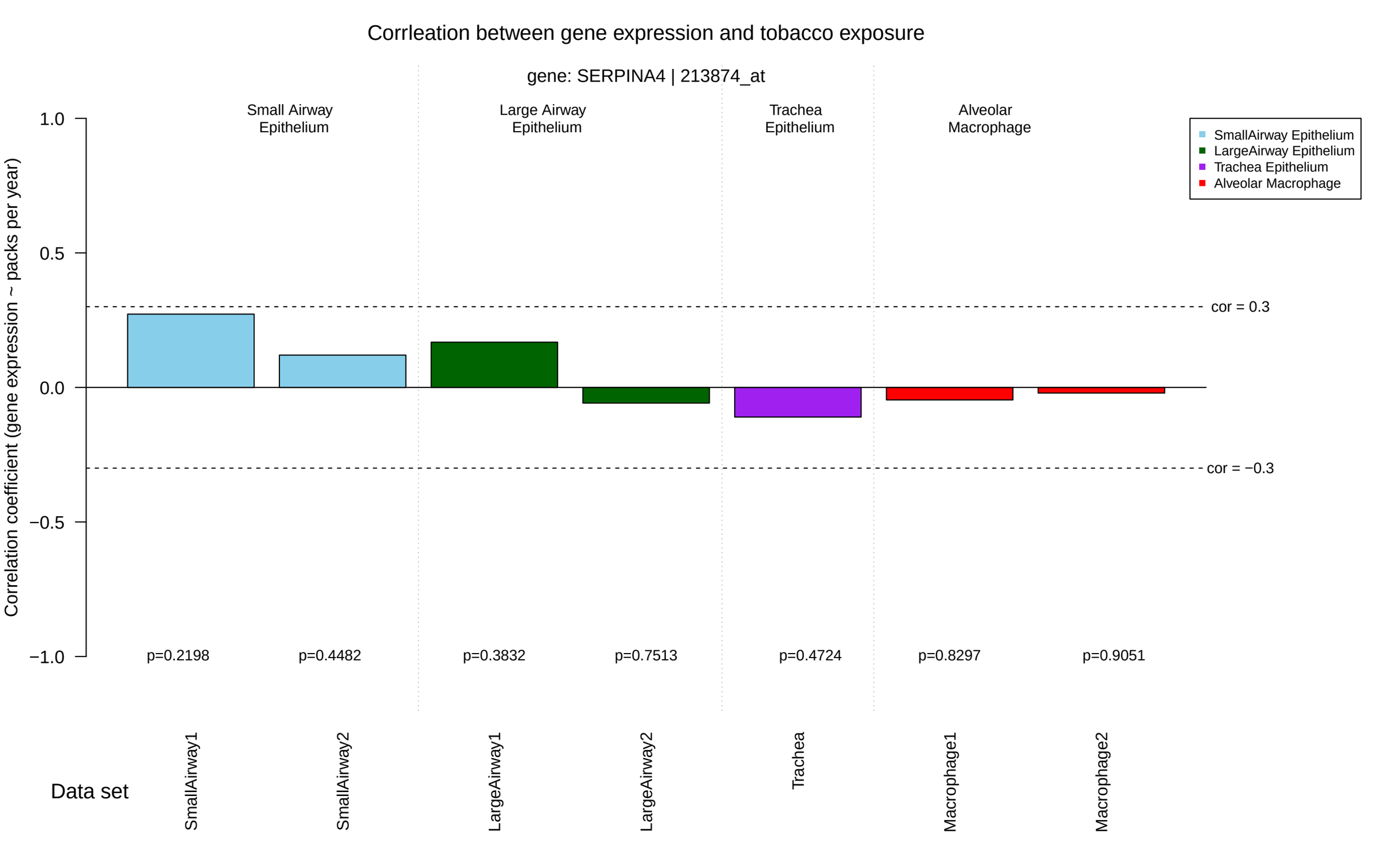

set: LargeAirway1 (p=0.0217)
SERPINA4 | 213874_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.336)
SERPINA4 | 213874_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4059)
SERPINA4 | 213874_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9697)
SERPINA4 | 213874_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7983)
SERPINA4 | 213874_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SERPINA4 | 213874_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2198
p=0.4482
p=0.3832
p=0.7513
p=0.4724
p=0.8297
p=0.9051
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2